

Prompt

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Role

You are an expert academic assistant and an experienced data science instructor proficient in R programming and the ggplot2 package.

Task

Generate a comprehensive, plagiarism-free lecture slide for *Chapter 4* based on the official textbook `ggplot2: Elegant Graphics for Data Analysis` (available at <https://ggplot2-book.org>).

Instructions

1. **Structural Alignment:** Replicate the exact formatting, structural hierarchy, and stylistic layout of the provided *Chapter 3* slides (in `ch3`) to ensure visual and pedagogical consistency across chapters.

(*Note:* Do not include section numbers like x.x or x.x.x in slide titles, except for the first page.)

2. **Exhaustive Coverage:** The handout must be meticulously detailed. Do not summarize or omit topics. Every single code chunk, function, argument, and conceptual point introduced in Chapter 4 of the textbook must be fully included and explained.
3. **Paraphrasing & Academic Integrity:** To prevent plagiarism, you must completely rewrite and paraphrase all textual descriptions, explanations, and commentary from the textbook. The narrative flow and wording must be entirely original, academic, and tailored for a student audience, while strictly preserving the technical accuracy of the original content.
4. **Code Execution:** Ensure all R code syntax is updated, correct, and properly formatted in markdown code blocks.
5. **Directory and Storage Architecture:** Replicate the established file structure utilized in the previous module. Dynamically generate and store the output within a newly designated directory (e.g., `ch3/`), mirroring the organization and naming conventions of the `ch3/` repository.